MR-PATH: A Latent Mixture Model for Heterogeneous Causal Mechanisms in Mendelian Randomization

Daniel Iong

University of Michigan, Ann Arbor

daniong@umich.edu

Joint work with Qingyuan Zhao, Yang Chen

Website: `http://danieliong.me/mr-path/`

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Outline

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2. Mechanistic Heterogeneity in MR
3. MR-PATH
   - Model Assumptions
   - Statistical inference
4. Results
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   - BMI-T2D
5. Conclusion

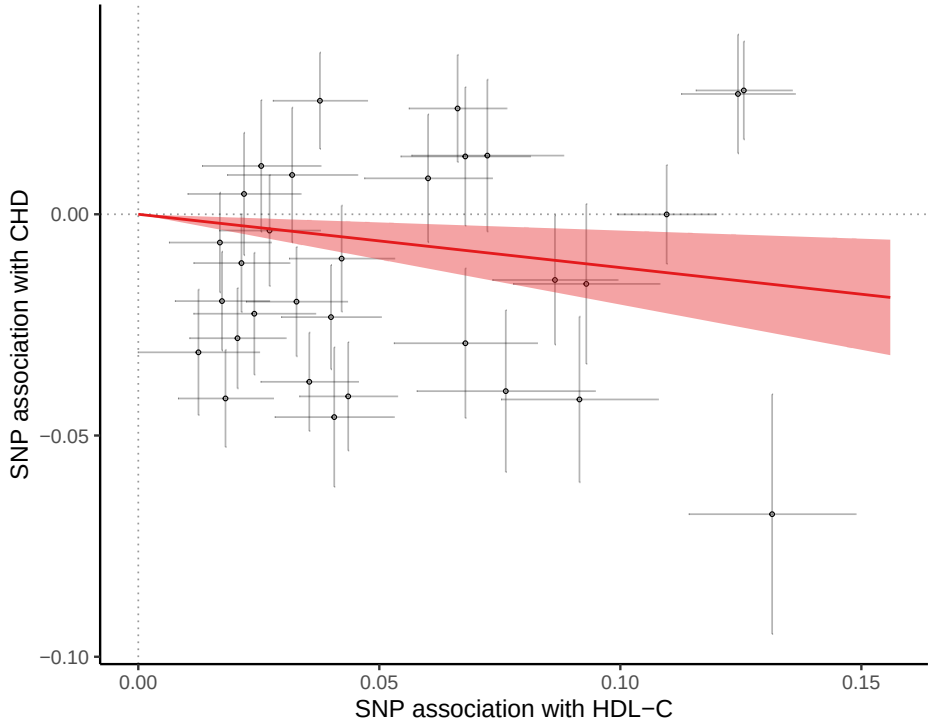
Motivation

- **Exclusion restriction assumption**: Instruments (genetic variants) can only affect the outcome through the risk exposure.
  - In MR, this assumption may be violated due to **pleiotropy**. Many pleiotropy-robust MR methods have been developed.
- Most robust MR methods rely on the **"effect homogeneity" assumption**: the risk exposure has the same causal effect for every individual.
- However, this assumption may be unrealistic in certain MR applications involving **multiple causal mechanisms**
  - e.g. the effect of HDL-cholesterol on coronary heart disease.

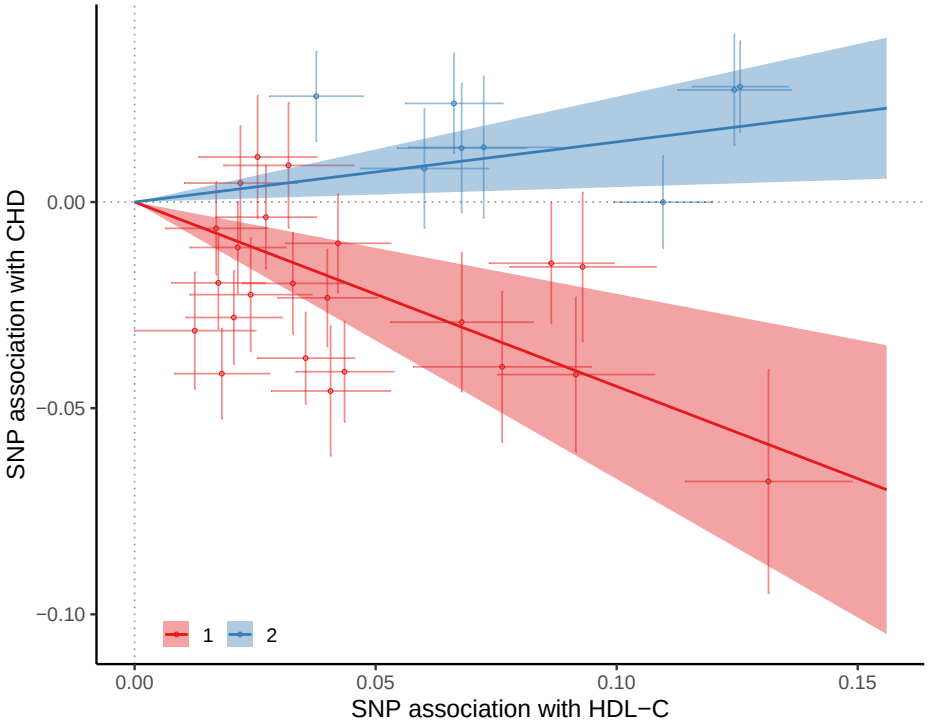
Our contributions

1. The concept of **mechanistic heterogeneity** in MR.
2. **MR-PATH**, a transparent mixture model to capture mechanistic heterogeneity.

Motivating examples: The effect of HDL cholesterol on coronary heart disease

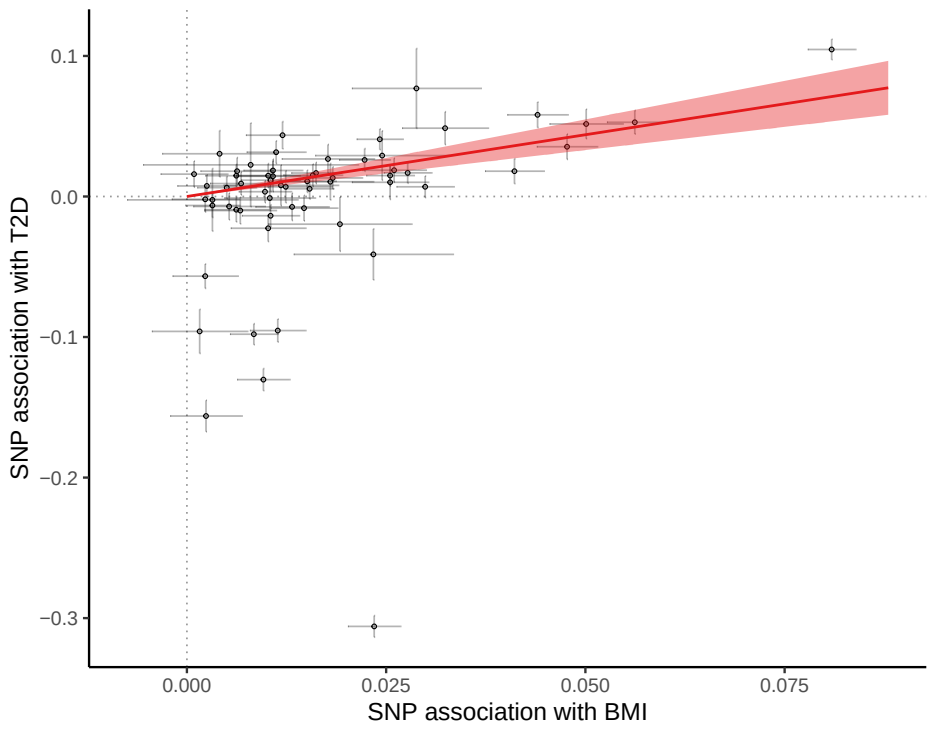


(a) MR-RAPS

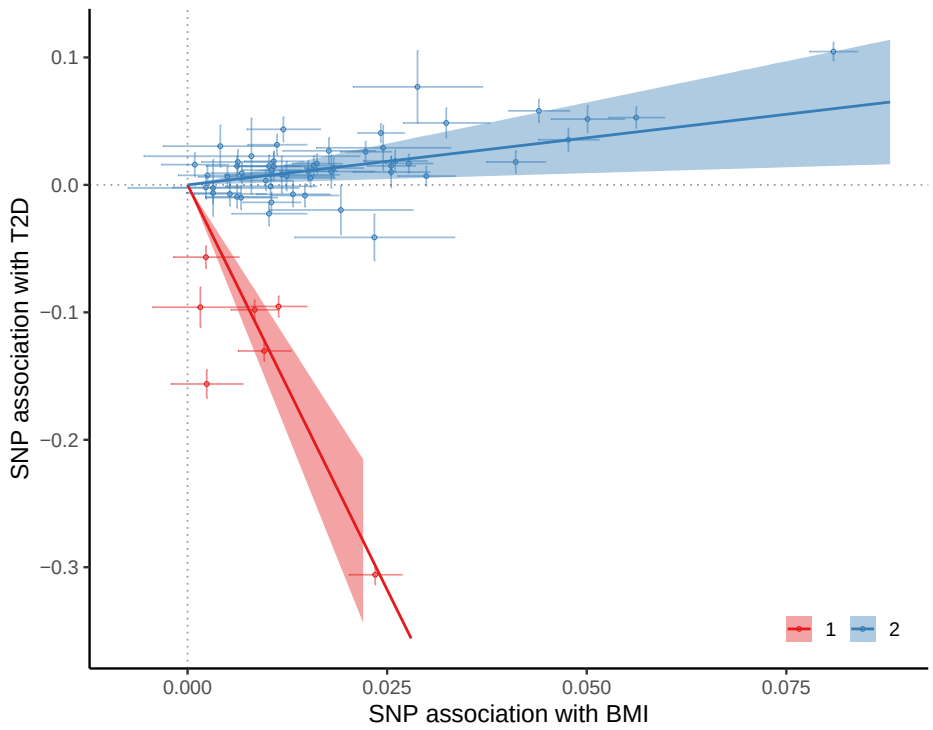


(b) MR-PATH

Motivating examples: The role of adiposity in type II diabetes

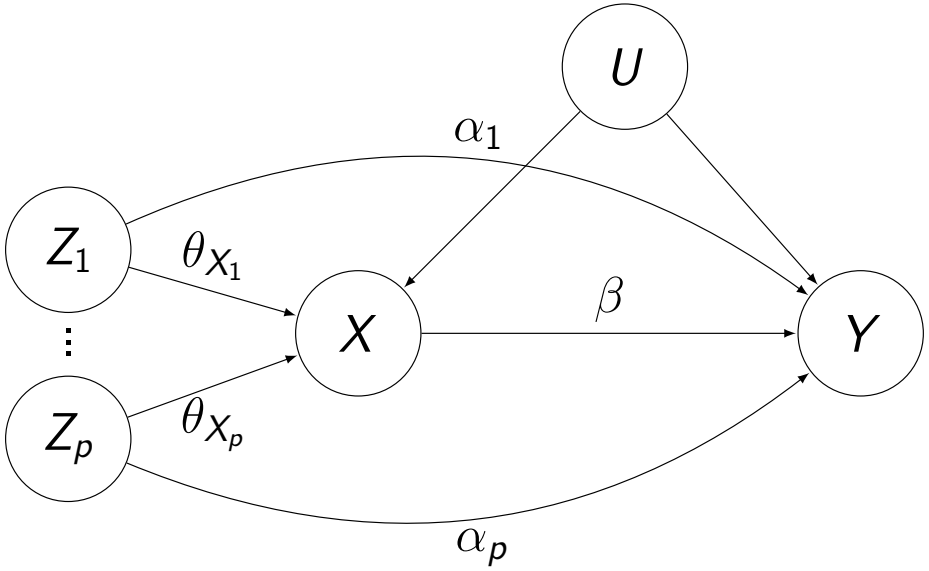


(a) MR-RAPS



(b) MR-PATH

Review: Linear structural equation model for MR



For exposure X, outcome Y, unobserved confounding variables U, and SNPs $Z_1, \ldots, Z_p$, the commonly assumed linear structural equation model is given by

$$X = \sum_{i=1}^{p} \theta_{X_i} Z_i + \eta_X U + E_X,$$

$$Y = \beta X + \sum_{i=1}^{p} \alpha_i Z_i + \eta_Y U + E_Y$$

Review: Linear structural equation model for MR

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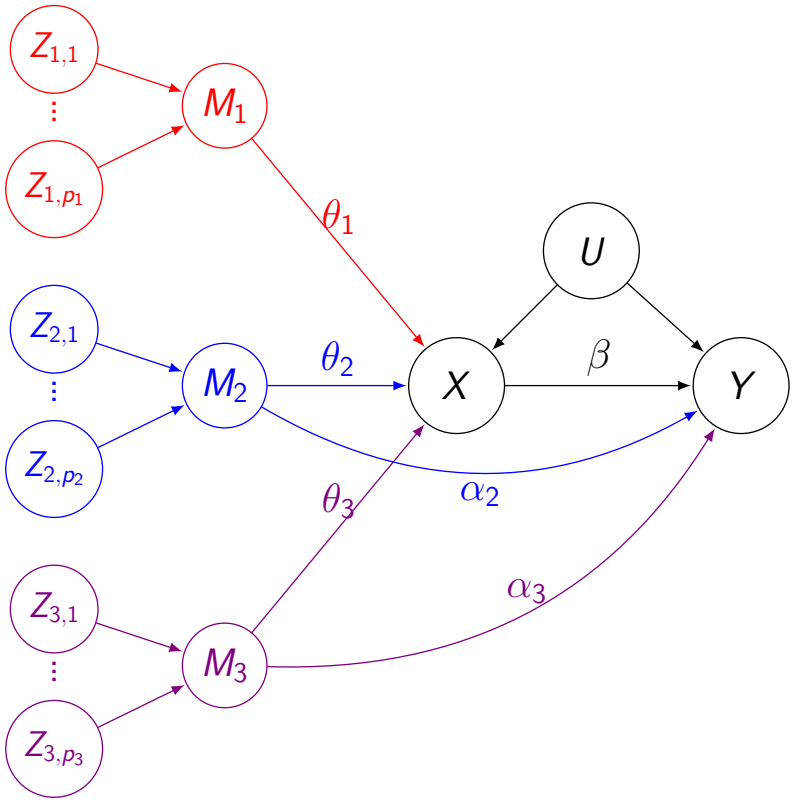
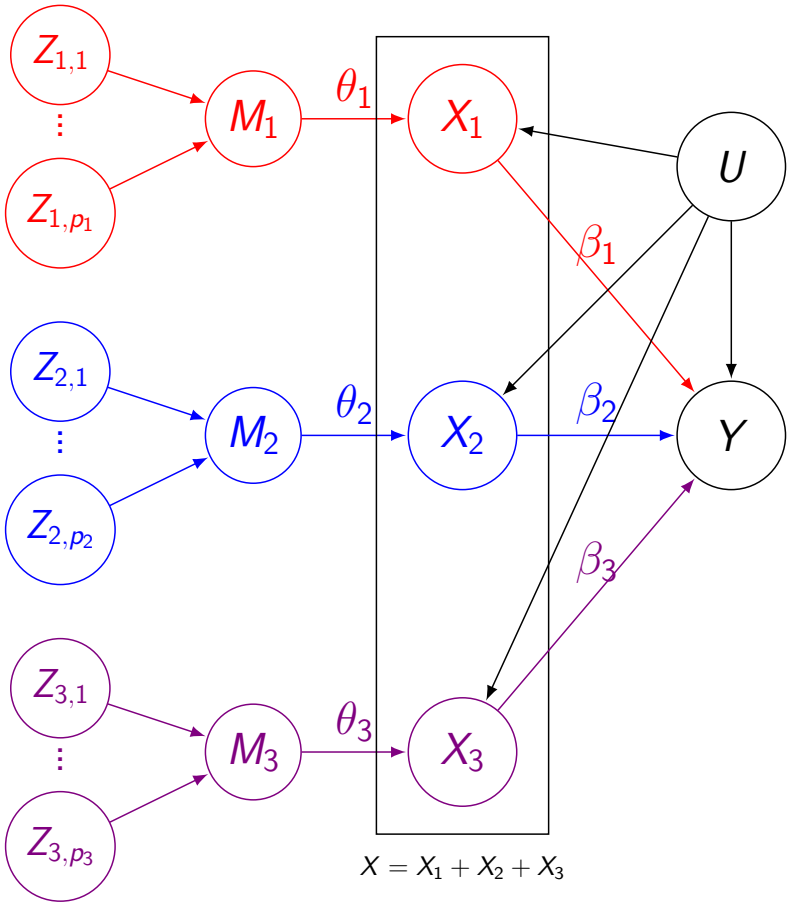
$$Y = \beta X + \sum_{i=1}^{p} \alpha_i Z_i + \eta_Y U + E_Y$$

- If Z_i is a valid instrument, $\theta_{X_i} \neq 0$, $Z_i \perp\!\!\!\perp \{U, E_X, E_Y\}$, and $\alpha_i = 0$.
- However, it is often the case that $\alpha_i \neq 0$ due to pleiotropy and multiple causal pathways.
- If $\alpha_i \neq 0$ for some SNPs, then the causal effect β cannot be estimated consistently without further assumptions on α_i.
  - e.g. $\alpha_i \sim N(0, \tau^2)$ for most SNPs.

Two scenarios of mechanistic heterogeneity



(a) Scenario 1: Multiple pathways of horizontal pleiotropy.



(b) Scenario 2: Multiple mechanisms for the exposure X.

Two scenarios of mechanistic heterogeneity

If we interpret the diagrams in the previous slide as linear structural equations as before, we can derive the Wald estimands for each instrument.

Instruments Z	**Pathway** M	**Effect of** M **on** X	**Effect of** M **on** Y	**Wald estimand**
		Scenario 1		
$Z_{1,1}, \ldots, Z_{1,p_1}$	M_1	θ_1	$\theta_1\beta$	β
$Z_{2,1}, \ldots, Z_{2,p_2}$	M_2	θ_2	$\theta_2\beta + \alpha_2$	$\beta + \alpha_2/\theta_2$
$Z_{3,1}, \ldots, Z_{3,p_3}$	M_3	θ_3	$\theta_3\beta + \alpha_3$	$\beta + \alpha_3/\theta_3$
		Scenario 2		
$Z_{1,1}, \ldots, Z_{1,p_1}$	M_1	θ_1	$\theta_1\beta_1$	β_1
$Z_{2,1}, \ldots, Z_{2,p_2}$	M_2	θ_2	$\theta_2\beta_2$	β_2
$Z_{3,1}, \ldots, Z_{3,p_3}$	M_3	θ_3	$\theta_3\beta_3$	β_3

- SNPs on the same pathway have the same Wald estimand, while SNPs across different pathways generally have different estimands.
- Mechanistic heterogeneity can arise even when all SNPs are valid instruments (Scenario 2).

MR-PATH: Model Assumptions

Assumption (Error-in-variables regression)

The observed SNP-exposure and SNP-outcome associations are distributed as

$$\begin{pmatrix} \hat{\theta}_{X_i} \\ \hat{\theta}_{Y_i} \end{pmatrix} \overset{indep.}{\sim} N\left(\begin{pmatrix} \theta_{X_i} \\ \beta_i \theta_{X_i} \end{pmatrix}, \begin{pmatrix} \sigma^2_{X_i} & 0 \\ 0 & \sigma^2_{Y_i} \end{pmatrix} \right), \quad i = 1, \ldots, p,$$

where σ_{X_i}, σ_{Y_i} are (fixed) measurement errors.

Assumption (Mixture model for mechanistic heterogeneity)

$$Z_i \sim Categorical\,(\pi_1, \ldots, \pi_K),$$
$$\beta_i | Z_i = k \sim N(\mu_k, \sigma^2_k), \quad k = 1, \ldots, K.$$

MR-PATH: Statistical Inference

1. **Monte-Carlo EM algorithm** for obtaining model parameter estimates
2. **Approximate confidence intervals** for quantifying uncertainty of the estimates
3. **Modified Bayesian Information criterion (BIC)** for selecting number of clusters

- We perform simulation studies to verify the efficacy of these inference procedures.
- See paper for implementation details.

Example: HDL-CHD

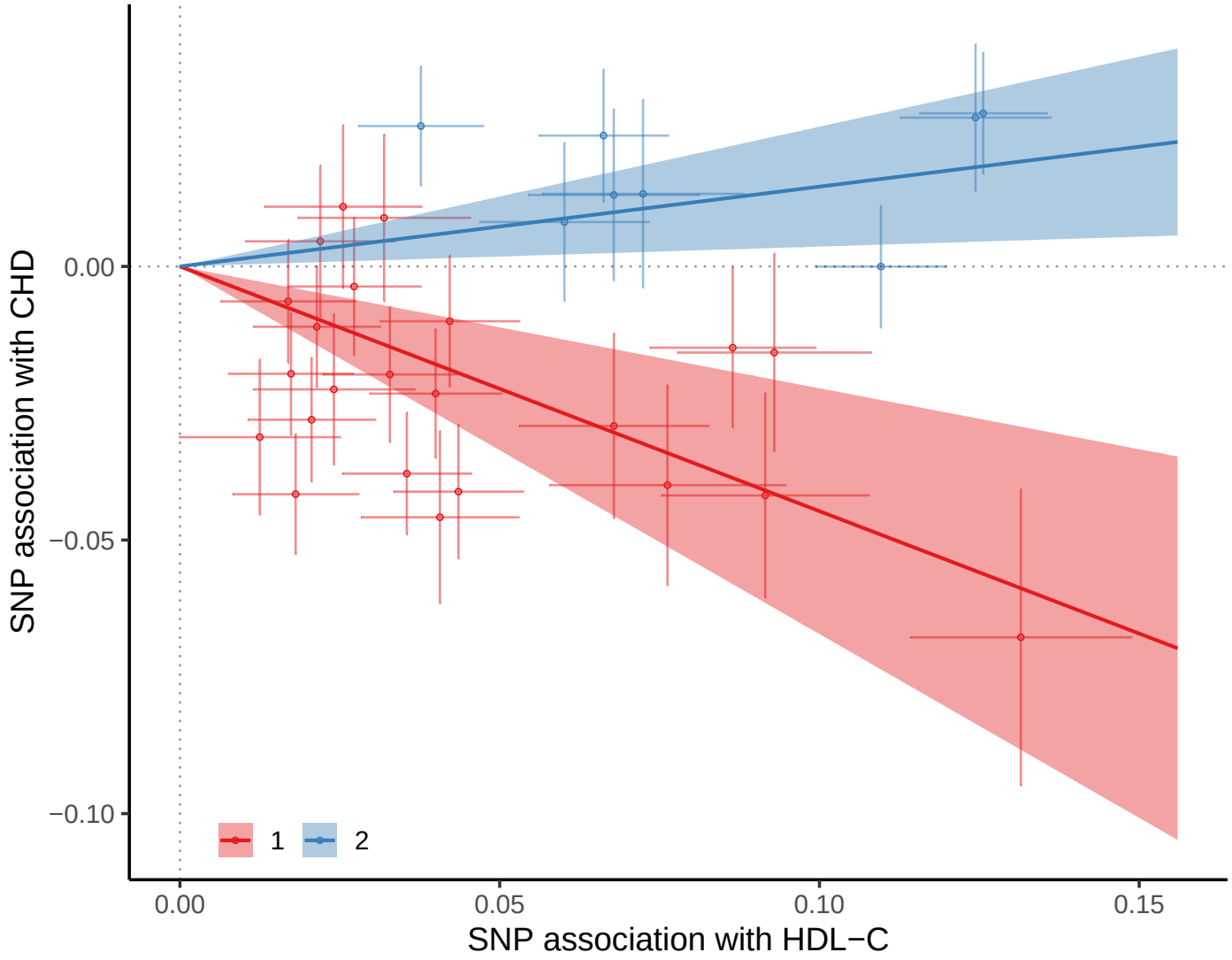
Data (Three-sample MR design)

- **Selection dataset**: **TODO**: Fill this in.
- **Exposure dataset**:
- **Outcome dataset**:

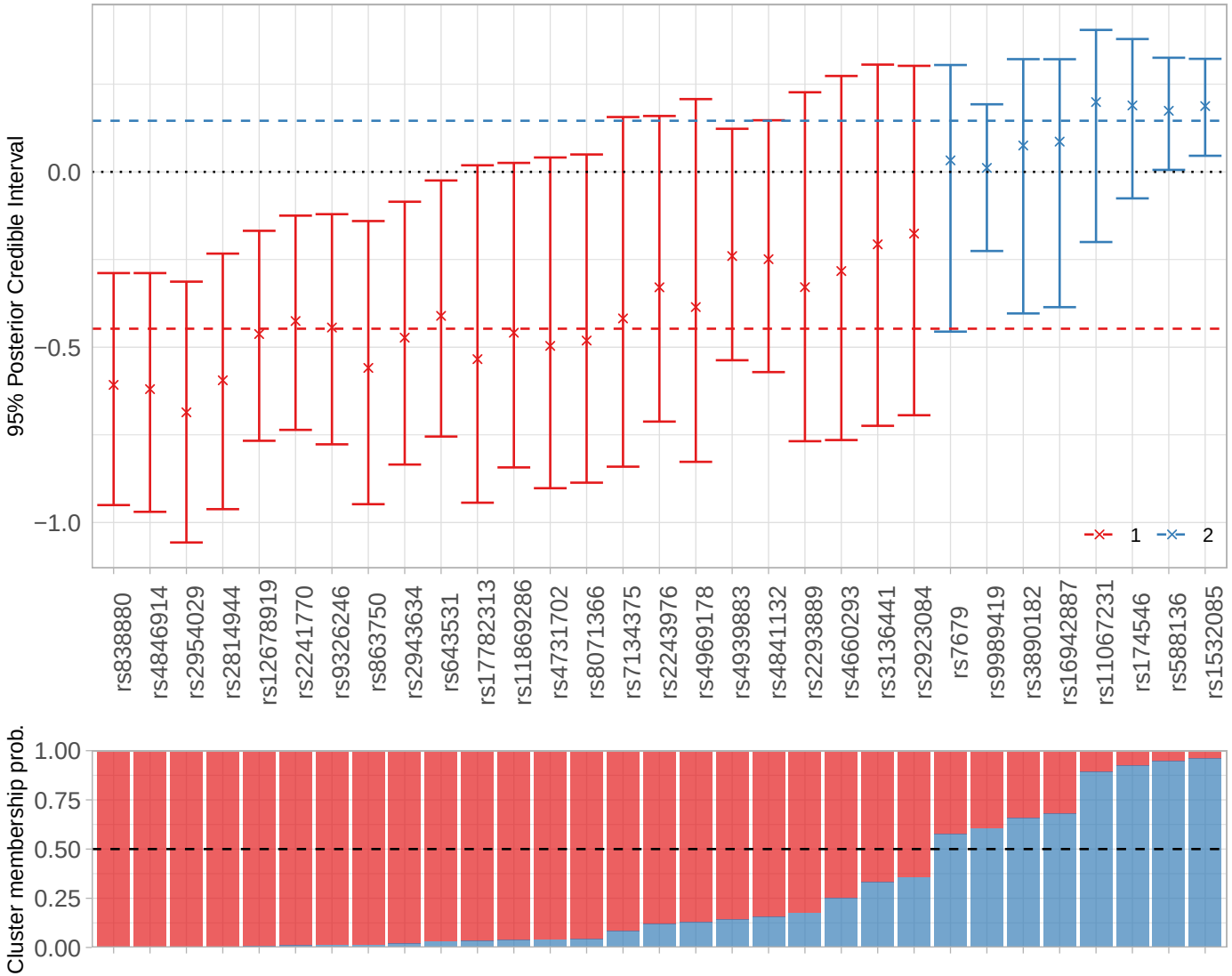
- We fit the MR-PATH model to this data.
- The modified BIC selects $K = 2$ clusters.

Example: HDL-CHD

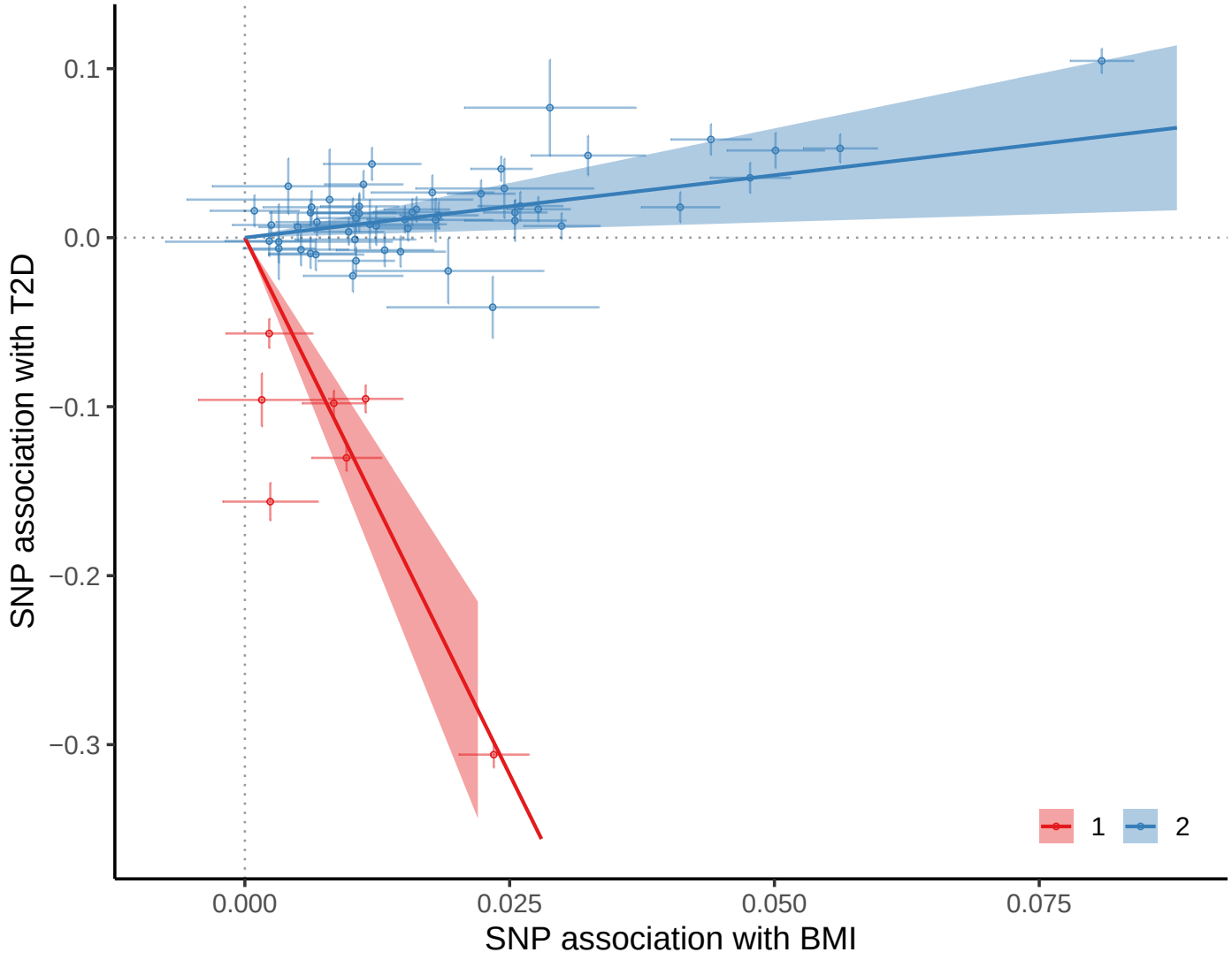
Interactive plots: `http://danieliong.me/mr-path/`.



Example: HDL-CHD



Example: BMI-T2D



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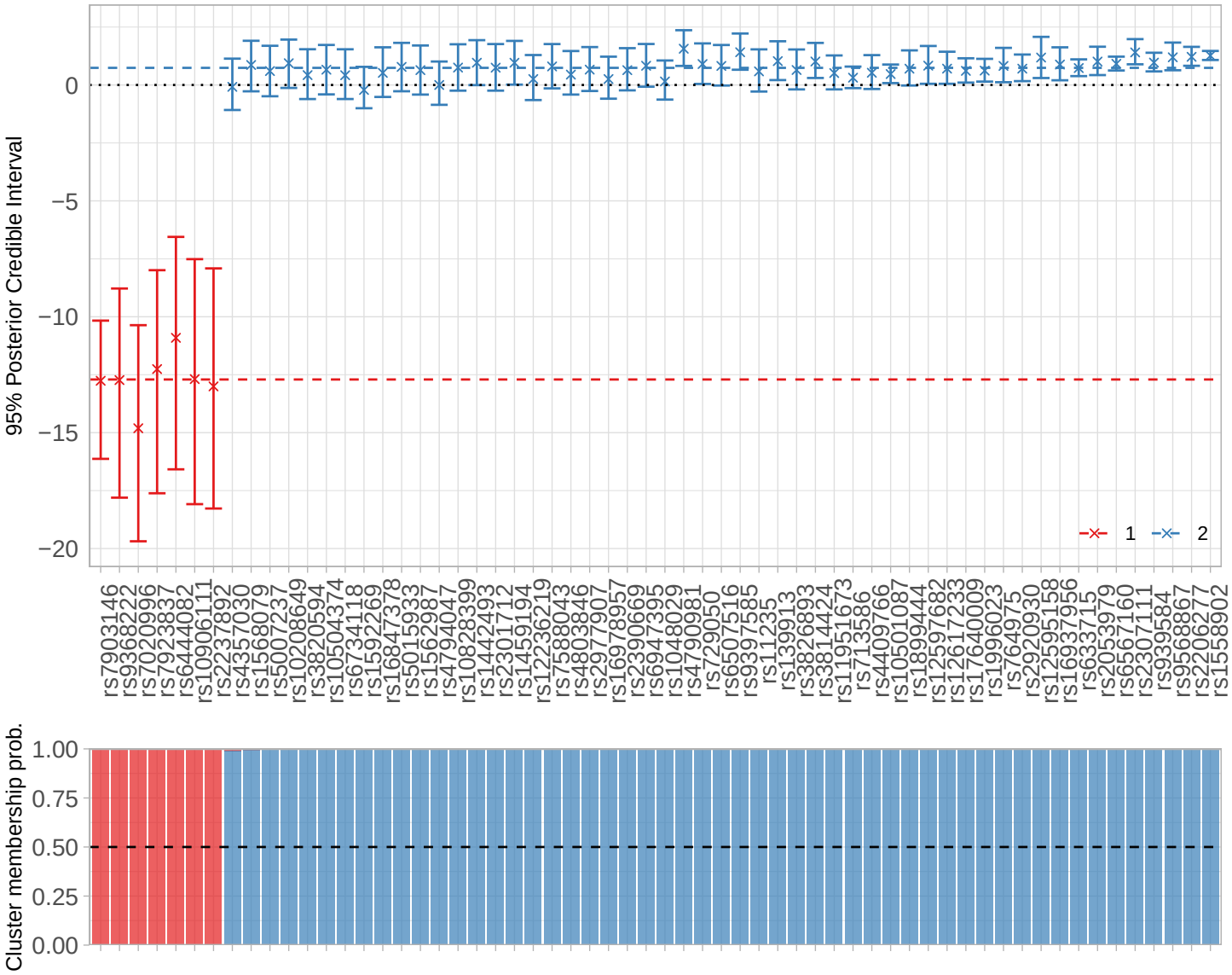

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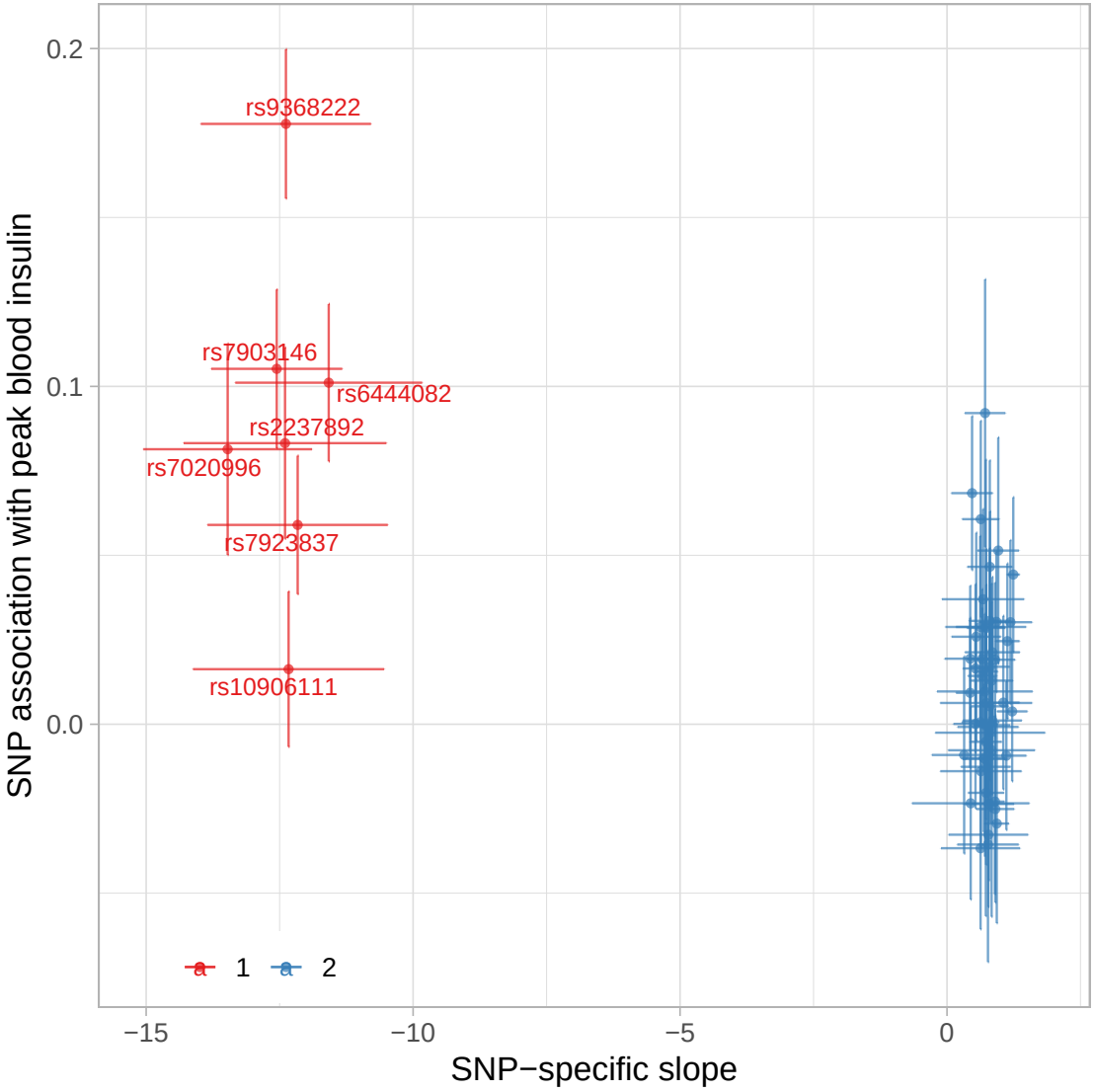

Concluding remarks

- There are a few other methods that relax the effect homogeneity assumption that we are aware of.
  - **MR-Clust**: Constructs mixture model based on SNP-specific Wald estimators.
  - **GRAPPLE**: A visualization tool that does not attempt to model different mechanisms explicitly.
  - **BESIDE-MR**: A Bayesian model averaging approach extends the profile likelihood used in MR RAPS.
- **Advantages of MR-PATH**
  - It does not require the individual instruments to be strong.
  - It's an interpretable generative model for multiple causal mechanisms which allows for many potential extensions (multivariate MR, correlated SNPs, etc).